Introduction to Time Series

➢ Purpose of the Notebook

- Demonstrates **time series attributes** using **synthetic data** in Python.
- Lets you **visually explore** how trend, seasonality, noise, and autocorrelation interact.

➢ Key Steps & Concepts

1. **Basic Time Series**
   - Even a **straight line** is a time series (time on x-axis, value on y-axis).
2. **Seasonality**
   - Regular **peaks & troughs** (repeating pattern).
   - Example: Store profits that dip when closed, peak after reopening, decay midweek, and rise again on weekends.
3. **Trend + Seasonality**
   - Pattern **increases over time** (e.g., growing business with regular cycles).
4. **Adding Noise**
   - Random variation hides patterns from the human eye but **ML can still detect them**.
5. **Autocorrelation**
   - Current value depends on previous values.
   - Different autocorrelation functions produce different decay/repeat effects.
   - Can cause **shrinking repeated patterns** or smooth decays.
6. **Impulses (Spikes)**
   - Sudden events (e.g., market shock).
   - With autocorrelation → spike followed by **decay curves**, possibly interrupted by new spikes.
7. **Impactful Event Simulation**
   - Create a series with seasonality that changes drastically after a single event (e.g., business failure, news shock).

➢ Key Takeaway:

By combining synthetic **trend, seasonality, noise, autocorrelation, and impulses**, you can simulate realistic scenarios to understand time series behavior before applying ML to **real-world data**.
